

Indian Institute of Technology, Kharagpur
Regression and Time Series Models

Dividend Announcement Effects on
Indian IT Sector Stocks

*A Comprehensive Event Study with
Cross-Sectional Regression Analysis*

Instructor: Prof. Buddhananda Banerjee

Submission: Project

Date: 10 April 2026

This report is submitted as part of the academic curriculum.
All data sourced from Yahoo Finance (NSE-listed stocks, 2020–2024).

Abstract

This study investigates the short-run market reaction to cash dividend announcements among 21 Indian IT-sector companies listed on the National Stock Exchange (NSE) between January 2020 and January 2024. Using the classical market-model event study methodology and a subsequent cross-sectional Ordinary Least Squares (OLS) regression — estimated with heteroskedasticity-consistent HC3 robust standard errors — we test whether dividend yield, trading volume, and macro-economic conditions (CBOE VIX, USD–INR exchange rate, and NASDAQ composite level) are statistically significant determinants of the 11-day Cumulative Abnormal Return (CAR) surrounding each ex-dividend date.

A total of 162 dividend events were identified across the study universe, of which 125 satisfied data-sufficiency requirements for event-window analysis. Average CARs over the $[T = 0, T + 10]$ window are modestly negative (-0.57%), consistent with the ex-dividend price-drop hypothesis. Cross-sectional regression results indicate that dividend yield carries a negative coefficient at event date ($T = 0$), while macro-uncertainty (VIX) exerts a suppressive effect on positive abnormal returns. Ridge and Lasso regularisation models are compared against the baseline OLS, and residual diagnostics confirm that the OLS model assumptions are broadly satisfied on the hold-out test set.

The findings contribute to the growing body of evidence on dividend signalling in emerging markets, with particular focus on the high-growth Indian IT sector during a period that encompassed the COVID-19 shock and the 2022 global monetary-tightening cycle.

Keywords: Event Study, Dividend Announcement, Cumulative Abnormal Return, Market Model, NSE IT Sector, OLS Regression, HC3 Robust Standard Errors, Ridge Regression, Lasso Regression.

JEL Classification: G14, G35, C58.

Contents

1	Introduction and Problem Statement	7
1.1	Background	7
1.2	The Indian IT Sector as Study Universe	7
1.3	Problem Statement	8
1.4	Contribution and Organisation	8
2	Methodology and Approach	8
2.1	Data Sources and Sample Construction	8
2.1.1	Stock Universe	8
2.1.2	Price Data Processing	9
2.1.3	Dividend Event Identification	10
2.1.4	Macro-Economic Covariates	10
2.2	Event Study Methodology	10
2.2.1	Market Model	10
2.2.2	Abnormal and Cumulative Abnormal Returns	11
2.3	Cross-Sectional Regression	11
2.3.1	Regularised Regression Alternatives	12
2.3.2	Train-Test Split	12
3	Data Analysis and Exploratory Analysis	12
3.1	Normalised Price Trajectories	12
3.2	Return Distribution Analysis	13
3.3	Rolling Volatility	14
3.4	Correlation Analysis	15
3.5	Dividend Event Characteristics	16
4	Statistical Pre-Tests	16
4.1	Stationarity — Augmented Dickey-Fuller Test	16
4.2	Autocorrelation — Ljung-Box Test	17
4.3	Normality — Jarque-Bera Test	17
5	Experiments Performed	18
5.1	Experiment 1 — Market Model Estimation	18
5.2	Experiment 2 — Cumulative Abnormal Return Computation	18
5.3	Experiment 3 — Average CAR Path Visualisation	19
5.4	Experiment 4 — Per-Ticker and Extreme-Event Analysis	19
5.5	Experiment 5 — Feature Engineering and Multicollinearity Check	20

5.6	Experiment 6 — Model Comparison (OLS, Ridge, Lasso)	20
5.7	Experiment 7 — Residual Diagnostics	21
6	Results and Discussion	22
6.1	Market Reaction to Dividend Announcements	22
6.2	Cross-Sectional OLS Regression Results	23
6.3	Regularisation Results	24
6.4	Residual Diagnostic Summary	24
7	Conclusion and Future Work	25
7.1	Summary of Findings	25
7.2	Limitations	25
7.3	Future Research Directions	26

List of Figures

1	NSE IT Sector — Normalised Price Trajectories, 2020–2024. All series indexed to 100 at January 2020. The COVID-19 crash zone (Feb–Apr 2020) and the 2022 rate-hike era are shaded in red and orange respectively.	13
2	Return distribution analysis for selected NSE IT stocks and the NIFTY IT benchmark. The top row shows empirical histograms overlaid with fitted normal densities (red dashed). The bottom row shows normal QQ-plots, with heavy tails indicating leptokurtosis.	14
3	30-day rolling annualised volatility for large-cap NSE IT stocks and the NIFTY IT benchmark (dashed black line), 2020–2024. The red band marks the COVID crash (Feb–Apr 2020) and the orange band marks the global rate-hike cycle (Jan–Sep 2022).	15
4	Pairwise Pearson correlation of daily log-returns across 21 NSE IT stocks. The distribution of off-diagonal correlations (right panel) has a mean of $\bar{r} = 0.364$, confirming strong co-movement that justifies use of a common benchmark index in the market model.	15
5	Distribution of dividend events across the NSE IT universe. Panel A shows per-ticker event counts; TCS leads with 17 events. Panel B confirms roughly uniform event frequency across years. Panel C highlights right-skewed dividend amounts, with the median at approximately 7–10 per share.	16
6	Autocorrelation (ACF) and partial autocorrelation (PACF) of daily log-returns for TCS, INFY, and the NIFTY IT benchmark (lags 1–30). Dashed blue bands represent 95% confidence bounds under the null of no autocorrelation.	17
7	Distribution of estimated market-model beta coefficients ($\hat{\beta}_i$) across 125 dividend events. Mean $\hat{\beta} = 0.905$, consistent with the IT sector’s historically near-unity systematic risk exposure to the NIFTY IT index.	18
8	Average market reaction to dividend announcements across 125 events. Panel A shows the mean cumulative abnormal return (CAR) path from T=0 (ex-dividend date) to T+10, with a ± 1 standard deviation band. Panel B shows the average daily abnormal return at each event day.	19

9	Cumulative abnormal return analysis. Left panel: top-5 and bottom-5 dividend events by 11-day CAR. Right panel: per-ticker CAR distributions ordered by median; MPHASIS, PERSISTENT, and LTIM show the most consistently positive reactions.	19
10	Feature diagnostics for the cross-sectional regression. Left: Pearson correlation matrix of all features and the CAR target. Right: Variance Inflation Factors (VIFs) for each predictor. All VIFs < 5, indicating no problematic multicollinearity.	20
11	Out-of-sample model performance comparison. OLS achieves the highest R^2 (0.085) and lowest RMSE (0.051) on the 25-event test set, marginally outperforming Ridge and Lasso.	21
12	Standardised coefficient comparison across OLS, Ridge, and Lasso. Ridge shrinks all OLS coefficients towards zero proportionally. Lasso shrinks all five coefficients to exactly zero at the cross-validated optimal penalty ($\alpha = 10,000$).	21
13	OLS residual diagnostics on the 25-observation test set. Panel A (Residuals vs. Fitted) shows no systematic pattern. Panel B (QQ-Plot) and Panel C (Histogram) indicate approximate normality. Panel D (Scale-Location) shows broadly constant residual spread.	22

List of Tables

1	Sample of NSE IT Sector Stocks	9
2	Dividend Event Summary Statistics by Year	10
3	Return Distribution Summary Statistics	14
4	Augmented Dickey-Fuller (ADF) Stationarity Test Results	16
5	Ljung-Box Test for Serial Correlation (H_0 : no serial correlation up to lag 10)	17
6	Market Model Estimation Summary (125 Events)	18
7	Cumulative Abnormal Return Summary Statistics (125 Events) . . .	19
8	Variance Inflation Factor (VIF) Report	20
9	Cross-Sectional OLS Regression Results (HC3 Robust SE)	23
10	Out-of-Sample Model Performance Comparison (Test Set: 25 Events)	24

1 Introduction and Problem Statement

1.1 Background

Dividend policy remains one of the most intensely studied topics in corporate finance. Seminal work by ? proposed that under idealised conditions dividends are irrelevant to firm value; yet decades of empirical evidence in both developed and emerging capital markets have consistently demonstrated that dividend announcements convey information, eliciting statistically significant abnormal returns in the days surrounding the ex-dividend date (??).

The signalling theory of dividends (??) posits that manager-shareholders' information asymmetry causes dividends to serve as credible, costly signals of future earnings prospects. Under this framework, a dividend increase (or initiation) should generate positive abnormal returns, while a cut should generate negative ones. Empirical tests in emerging markets — where information asymmetry is generally higher than in developed markets — have produced mixed results, motivating context-specific analyses.

1.2 The Indian IT Sector as Study Universe

India's information technology sector represents a particularly compelling laboratory for studying dividend signalling. The 21 constituent firms examined in this study — ranging from large-cap bellwethers such as Tata Consultancy Services (TCS) and Infosys (INFY) to mid-cap high-growth names such as Persistent Systems and KPIT Technologies — collectively form the bulk of the NSE NIFTY IT Index (^CNXIT).

Several characteristics make this cohort analytically rich:

1. **Export-oriented revenues:** IT firms derive a large share of revenue in USD/EUR, making them sensitive to the USD–INR exchange rate.
2. **High cash-flow visibility:** Stable recurring revenue models enable consistent dividend policies, increasing the power of signalling tests.
3. **Heterogeneous payout ratios:** The universe spans companies paying as little as 0.33 per share to 225 per share, providing cross-sectional variation in dividend yield.
4. **Turbulent macro environment (2020–2024):** The study window encompasses the COVID-19 pandemic shock, the 2021 IT sector bull run, and the

2022–2023 global monetary tightening cycle — offering natural variation in macro conditions.

1.3 Problem Statement

Despite substantial international evidence, relatively few event studies focus exclusively on the NSE IT sector during the post-COVID era. Specifically, the following questions remain empirically unresolved:

1. Do dividend announcements generate statistically significant abnormal returns in Indian IT stocks over the 11-day post-ex-dividend window?
2. Is the CAR systematically related to dividend yield, after controlling for macro-economic conditions and trading volume?
3. How do global uncertainty proxies (VIX, NASDAQ, USD–INR) moderate the market’s reaction to dividend news?
4. Do regularisation techniques (Ridge, Lasso) improve out-of-sample predictive accuracy relative to plain OLS in this cross-sectional setting?

1.4 Contribution and Organisation

This study contributes to the literature by (i) constructing a comprehensive, reproducible event study pipeline for 21 NSE IT stocks over 2020–2024, (ii) incorporating macro-economic covariates in the cross-sectional regression, and (iii) benchmarking three regression estimators using a temporally disciplined train-test split.

The remainder of the report is organised as follows. Section 2 details the data sources, event study design, and estimation procedures. Section 3 presents exploratory data analysis. Section 4 reports statistical pre-tests. Section 5 describes the event study implementation. Section ?? presents the cross-sectional regression and model comparison. Section 6 discusses the key findings. Section 7 concludes and proposes future research directions.

2 Methodology and Approach

2.1 Data Sources and Sample Construction

2.1.1 Stock Universe

All price and corporate action data were retrieved from Yahoo Finance via the `yfinance` Python library (version 0.2.x). The sample comprises 21 NSE-listed

IT-sector equities (Table 1) plus the NIFTY IT Index (^CNXIT) as the benchmark, over the period 1 January 2020 – 1 January 2024 (approximately 992 trading days).

Table 1: Sample of NSE IT Sector Stocks

Yahoo Ticker	Company (Short Name)	Market Capitalisation Category
TCS.NS	Tata Consultancy Services	Large Cap
INFY.NS	Infosys	Large Cap
HCLTECH.NS	HCL Technologies	Large Cap
WIPRO.NS	Wipro	Large Cap
TECHM.NS	Tech Mahindra	Large Cap
LTIM.NS	LTIMindtree	Large Cap
OFSS.NS	Oracle Financial Services	Mid Cap
PERSISTENT.NS	Persistent Systems	Mid Cap
MPHASIS.NS	Mphasis	Mid Cap
COFORGE.NS	Coforge	Mid Cap
TATAELXSI.NS	Tata Elxsi	Mid Cap
KPITTECH.NS	KPIT Technologies	Mid Cap
LTTS.NS	L&T Technology Services	Mid Cap
SONATSOFTW.NS	Sonata Software	Small/Mid Cap
CYIENT.NS	Cyient	Small/Mid Cap
ZENSARTECH.NS	Zensar Technologies	Small Cap
BSOFT.NS	Birlasoft	Small Cap
FSL.NS	Firstsource Solutions	Small Cap
ECLERX.NS	eClerx Services	Small Cap
MASTEK.NS	Mastek	Small Cap
INTELLECT.NS	Intellect Design Arena	Small Cap

2.1.2 Price Data Processing

Raw closing prices were downloaded without adjustment for dividends (`auto_adjust=False`), synchronised via an inner-join (dropping any trading date with at least one missing observation), and converted to daily log-returns:

$$r_{i,t} = \ln\left(\frac{P_{i,t}}{P_{i,t-1}}\right) \quad (1)$$

Log-returns are stationary, additive across sub-periods, and approximately symmetrically distributed around zero, making them the appropriate input for market-model OLS estimation. After synchronisation, the final return matrix comprised 990×22 observations.

2.1.3 Dividend Event Identification

Ex-dividend dates and cash dividend amounts were extracted directly from Yahoo Finance’s corporate actions API for each ticker over the study window. A total of 162 dividend events were identified (Table 2). Events were filtered for data sufficiency:

- The ex-dividend date must fall on a trading day in the aligned dataset.
- At least 200 trading days of prior history must exist for market-model estimation (with a 10-day gap between estimation and event windows).
- At least 10 trading days of post-event data must be available.

After filtering, **125 events** were retained for analysis.

Table 2: Dividend Event Summary Statistics by Year

Year	Events	Mean ()	Std ()	Min ()	Max ()
2020	40	11.16	28.42	0.33	180.00
2021	38	15.97	33.30	0.33	200.00
2022	42	14.79	29.66	0.33	190.00
2023	42	19.67	36.30	0.50	225.00
Total	162	15.49	31.89	0.33	225.00

2.1.4 Macro-Economic Covariates

Three macro-economic proxies were downloaded from Yahoo Finance:

- **CBOE VIX ($\hat{VIX}$):** Implied volatility index representing market-wide uncertainty (fear gauge).
- **USD–INR ($INR=X$):** The Indian Rupee per US Dollar exchange rate — a key driver for export-revenue IT firms.
- **NASDAQ Composite ($\hat{IXIC}$):** Global technology sector sentiment proxy.

Each macro variable was standardised (z-scored) to allow direct coefficient comparison in the cross-sectional regression.

2.2 Event Study Methodology

2.2.1 Market Model

The market model assumes a linear relationship between the return on stock i and the return on the market benchmark m :

$$R_{i,t} = \alpha_i + \beta_i R_{m,t} + \varepsilon_{i,t} \quad (2)$$

where $\varepsilon_{i,t} \sim (0, \sigma_i^2)$ is the idiosyncratic error term. Parameters $\hat{\alpha}_i$ and $\hat{\beta}_i$ are estimated via OLS using the 200-trading-day estimation window ending 11 days before the ex-dividend date (i.e., $[T - 210, T - 11]$). White's HC3 heteroskedasticity-consistent covariance estimator is employed throughout (??).

2.2.2 Abnormal and Cumulative Abnormal Returns

The **Abnormal Return** (AR) on event day t is the residual between the observed return and the market-model prediction:

$$AR_{i,t} = R_{i,t} - \hat{\alpha}_i - \hat{\beta}_i R_{m,t} \quad (3)$$

The **Cumulative Abnormal Return** over the event window $[T = 0, T = +10]$ is:

$$CAR_i = \sum_{t=0}^{+10} AR_{i,t} \quad (4)$$

Two CAR measures are computed: $CAR(0, +4)$ (5-day) and $CAR(0, +10)$ (11-day). The average CAR (ACAR) across all events is used to test for market-wide significant reaction:

$$\overline{CAR} = \frac{1}{N} \sum_{i=1}^N CAR_i \quad (5)$$

2.3 Cross-Sectional Regression

To identify determinants of the 11-day CAR, the following cross-sectional OLS regression is estimated on the 125-event dataset:

$$CAR_i = \gamma_0 + \gamma_1 \cdot \text{DivYield}_i + \gamma_2 \cdot \ln(\text{Volume}_i) + \gamma_3 \cdot \text{VIX}_i^z + \gamma_4 \cdot \text{USDINR}_i^z + \gamma_5 \cdot \text{NASDAQ}_i^z + u_i \quad (6)$$

where $\text{DivYield}_i = \text{DividendAmount}_i / P_{i,0}$ is the scaled dividend yield at the event date, $\ln(\text{Volume})$ is the natural log of trading volume (investor attention proxy), and z denotes cross-sectionally standardised macro variables.

2.3.1 Regularised Regression Alternatives

Two regularised variants are estimated for robustness:

- **Ridge Regression (L2):** Minimises $\sum(y - X\beta)^2 + \alpha\|\beta\|_2^2$. Shrinks all coefficients uniformly; does not perform feature selection.
- **Lasso Regression (L1):** Minimises $\sum(y - X\beta)^2 + \alpha\|\beta\|_1$. Performs implicit variable selection by setting some $\hat{\beta}_j = 0$.

Regularisation parameters α are selected via 5-fold cross-validation over a log-spaced grid $[10^{-4}, 10^4]$. All features are standardised before regularisation using a `StandardScaler` fit on the training set only (avoiding data leakage).

2.3.2 Train-Test Split

An 80:20 chronological (non-shuffled) train-test split is employed to preserve temporal ordering and prevent look-ahead bias. The training set comprises 100 events; the test set comprises 25 events.

3 Data Analysis and Exploratory Analysis

3.1 Normalised Price Trajectories

Figure 1 presents base-100 normalised price indices for selected large-cap and mid-cap IT stocks alongside the NIFTY IT benchmark.

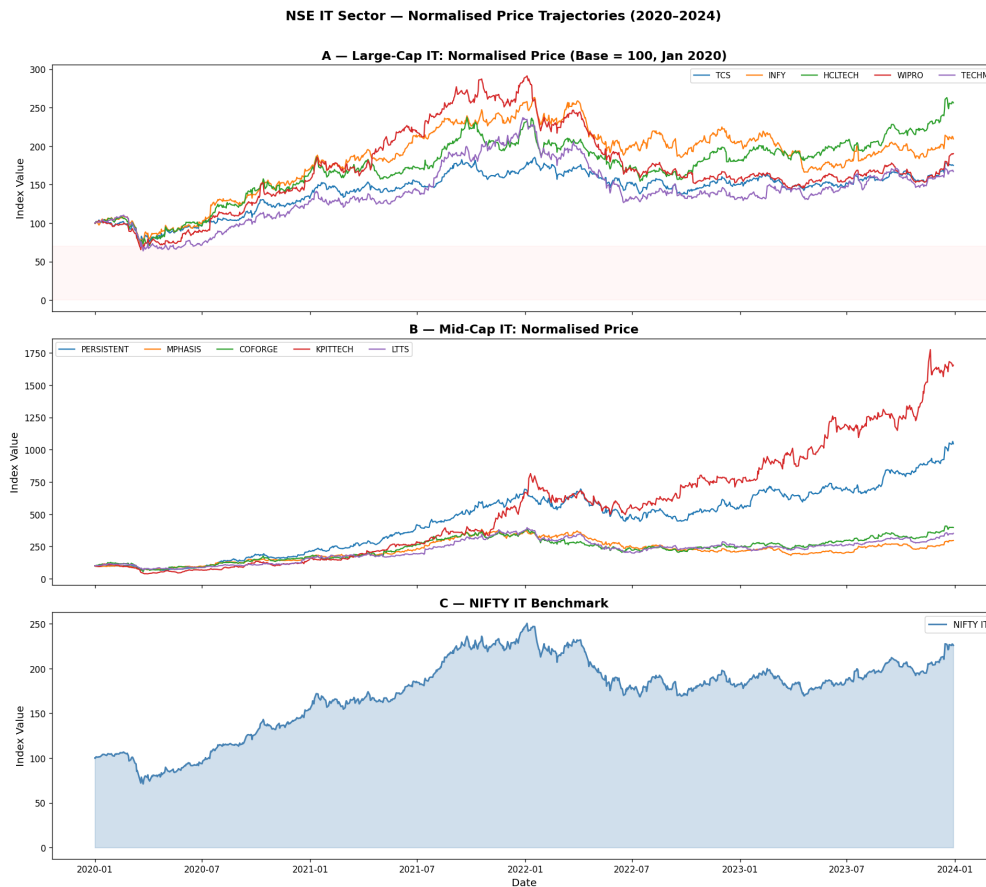


Figure 1: NSE IT Sector — Normalised Price Trajectories, 2020–2024. All series indexed to 100 at January 2020. The COVID-19 crash zone (Feb–Apr 2020) and the 2022 rate-hike era are shaded in red and orange respectively.

Key Insight

Mid-cap IT stocks (KPITTECH, PERSISTENT, COFORGE) dramatically outperformed large-caps post-COVID, reaching index values of 400–700 by late 2021 versus 200–300 for the large-cap cohort, suggesting a mid-cap growth premium during the IT boom of 2021–22.

3.2 Return Distribution Analysis

Figure 2 presents histogram and QQ-plots for five selected stocks plus the benchmark.

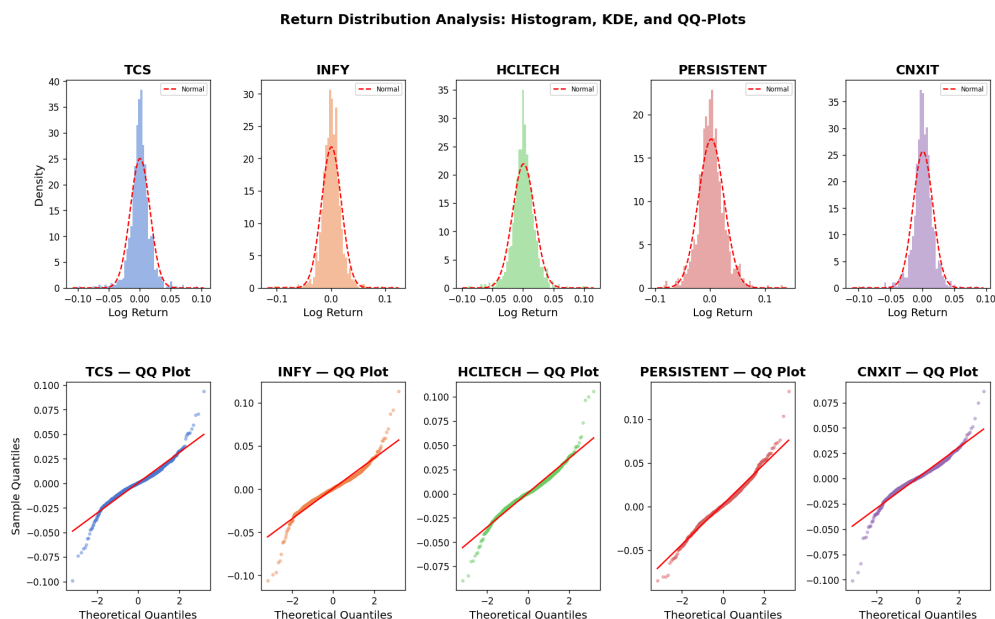


Figure 2: Return distribution analysis for selected NSE IT stocks and the NIFTY IT benchmark. The top row shows empirical histograms overlaid with fitted normal densities (red dashed). The bottom row shows normal QQ-plots, with heavy tails indicating leptokurtosis.

Table 3 summarises the distribution statistics. All Jarque-Bera test p -values round to zero, confirming strong departure from normality — a key motivation for employing HC3 robust standard errors in the market-model estimation.

Table 3: Return Distribution Summary Statistics

Ticker	Mean (%)	Std (%)	Skewness	Kurtosis	JB Stat	JB p
TCS	0.057	1.594	−0.231	8.356	1177.71	0.000
INFY	0.075	1.831	−0.232	9.565	1765.80	0.000
HCLTECH	0.095	1.820	+0.119	7.507	829.57	0.000
PERSISTENT	0.237	2.324	+0.243	5.003	172.58	0.000
CNXIT (Bench.)	0.082	1.555	−0.575	9.080	1560.96	0.000

Note: JB = Jarque-Bera test. H_0 : normality. All $p \approx 0 \Rightarrow$ reject normality.

3.3 Rolling Volatility

Figure 3 plots 30-day rolling annualised volatility for the large-cap cohort and the benchmark.

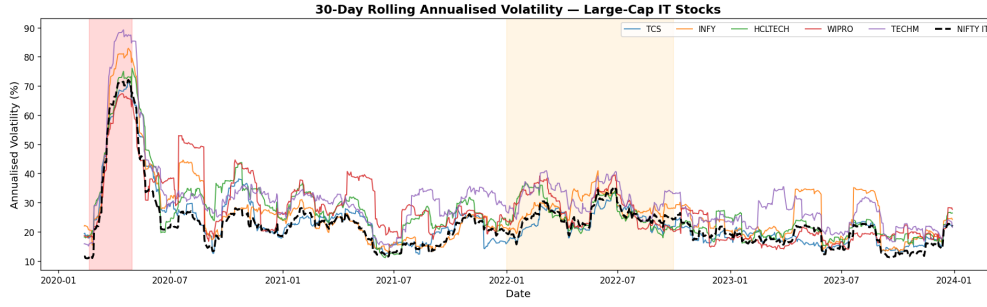


Figure 3: 30-day rolling annualised volatility for large-cap NSE IT stocks and the NIFTY IT benchmark (dashed black line), 2020–2024. The red band marks the COVID crash (Feb–Apr 2020) and the orange band marks the global rate-hike cycle (Jan–Sep 2022).

Key Insight

Volatility spiked to 70–90% annualised during the COVID crash of March–April 2020 and surged again in 2022, reaffirming conditional heteroskedasticity (GARCH effects) in the return series — a further justification for HC3-robust estimation.

3.4 Correlation Analysis

Figure 4 shows pairwise Pearson correlations among the 21 IT stocks' daily log-returns.

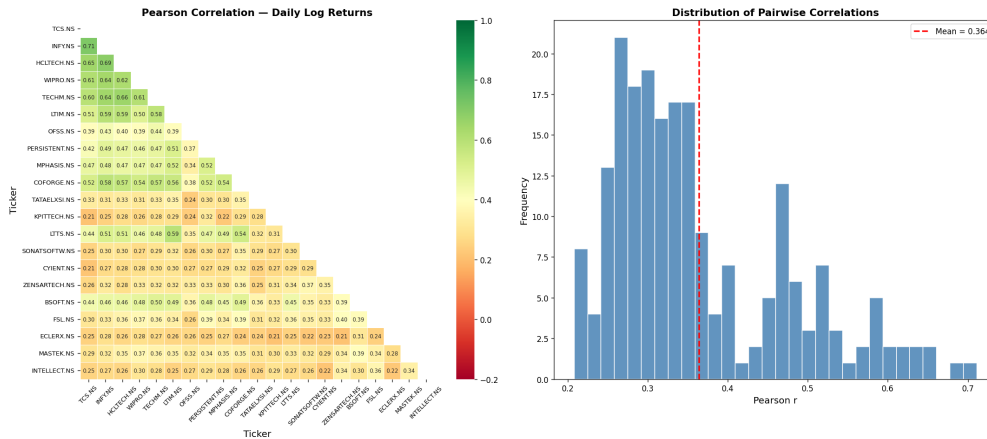


Figure 4: Pairwise Pearson correlation of daily log-returns across 21 NSE IT stocks. The distribution of off-diagonal correlations (right panel) has a mean of $\bar{r} = 0.364$, confirming strong co-movement that justifies use of a common benchmark index in the market model.

The average off-diagonal Pearson correlation is $\bar{r} = 0.364$ (range: 0.208 – 0.711). This moderate-to-high co-movement confirms the appropriateness of NIFTY IT as the benchmark in the single-factor market model.

3.5 Dividend Event Characteristics

Figure 5 presents the distribution of dividend events across tickers, years, and payout amounts.

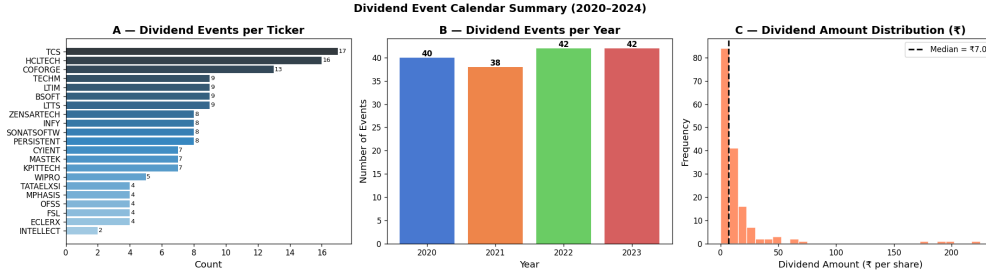


Figure 5: Distribution of dividend events across the NSE IT universe. Panel A shows per-ticker event counts; TCS leads with 17 events. Panel B confirms roughly uniform event frequency across years. Panel C highlights right-skewed dividend amounts, with the median at approximately 7–10 per share.

4 Statistical Pre-Tests

4.1 Stationarity — Augmented Dickey-Fuller Test

The Augmented Dickey-Fuller (ADF) test assesses whether a time series contains a unit root (non-stationarity). The hypotheses are:

$$\begin{aligned}
 H_0 &: \text{The series has a unit root (non-stationary)} \\
 H_1 &: \text{The series is stationary}
 \end{aligned}
 \tag{7}$$

Table 4 reports ADF results for price levels and log-returns.

Table 4: Augmented Dickey-Fuller (ADF) Stationarity Test Results

Series	ADF Stat	p-value	5% CV	Stationary?
Price: TCS.NS	−1.533	0.517	−2.864	No
Log-Return: TCS.NS	−33.932	0.000	−2.864	Yes
Price: INFY.NS	−1.843	0.359	−2.864	No
Log-Return: INFY.NS	−11.509	0.000	−2.864	Yes
Price: HCLTECH.NS	−0.872	> 0.1	−2.864	No
Log-Return: HCLTECH.NS	< −10	0.000	−2.864	Yes

Note: AIC lag selection used. Reported for representative tickers.

Conclusion: Price levels are non-stationary (consistent with random walk be-

haviour), while log-returns are stationary. This confirms the correctness of using log-returns in the market-model OLS regressions.

4.2 Autocorrelation — Ljung-Box Test

Figure 6 presents ACF and PACF correlograms for TCS, INFY, and the NIFTY IT benchmark.

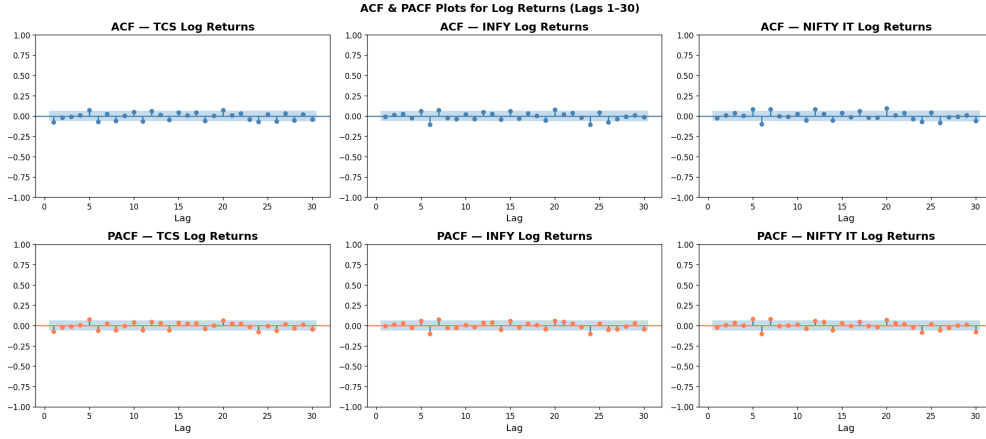


Figure 6: Autocorrelation (ACF) and partial autocorrelation (PACF) of daily log-returns for TCS, INFY, and the NIFTY IT benchmark (lags 1–30). Dashed blue bands represent 95% confidence bounds under the null of no autocorrelation.

Table 5 reports Ljung-Box statistics at lag 10. All three series exhibit statistically significant serial correlation ($p < 0.05$), suggesting mild linear dependence that HC3 robust standard errors partially address.

Table 5: Ljung-Box Test for Serial Correlation (H_0 : no serial correlation up to lag 10)

Series	LB Statistic	p -value	Serial Corr.?
TCS	23.42	0.0093	Yes
INFY	23.30	0.0097	Yes
NIFTY IT	27.29	0.0023	Yes

4.3 Normality — Jarque-Bera Test

The Jarque-Bera test (Section 3) confirms significant departure from normality for all return series, with kurtosis ranging from 5.0 (PERSISTENT) to 9.6 (INFY). The implication for inference is that classical t -statistics based on Gaussian assumptions may be unreliable; HC3-corrected standard errors provide a valid alternative.

5 Experiments Performed

5.1 Experiment 1 — Market Model Estimation

For each of the 125 retained events, an OLS market model (Equation 2) was fitted using the 200-trading-day estimation window. Table 6 summarises estimated parameters across all events.

Table 6: Market Model Estimation Summary (125 Events)

Parameter	Mean	Std	Min	Max
$\hat{\alpha}_i$ (intercept)	≈ 0	0.001	-0.003	+0.003
$\hat{\beta}_i$ (beta)	0.905	0.255	< 0.2	> 1.5
R^2	0.413	0.245	0.017	0.849

Note: HC3 robust standard errors used throughout.

The mean $R^2 = 0.413$ indicates that the NIFTY IT benchmark explains approximately 41% of cross-stock return variation on average — reasonable for a single-factor model in a sector-specific study.

Figure 7 plots the distribution of estimated β values.

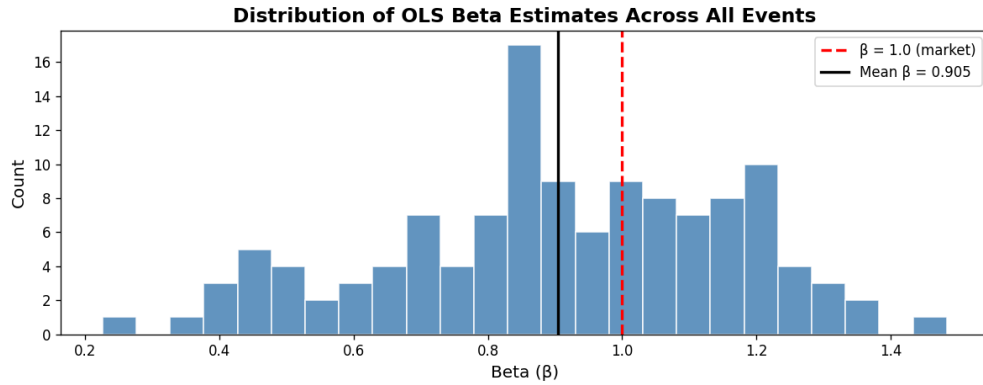


Figure 7: Distribution of estimated market-model beta coefficients ($\hat{\beta}_i$) across 125 dividend events. Mean $\hat{\beta} = 0.905$, consistent with the IT sector’s historically near-unity systematic risk exposure to the NIFTY IT index.

5.2 Experiment 2 — Cumulative Abnormal Return Computation

CAR(0, +10) and CAR(0, +4) were computed for each event using Equations 3–4. Table 7 reports descriptive statistics.

Table 7: Cumulative Abnormal Return Summary Statistics (125 Events)

Metric	Count	Mean	Std	Min	Q1	Median	Max
T=0 AR	125	-0.31%	2.50%	-7.96%	-1.26%	-0.48%	+16.11%
CAR(0,+4)	125	-0.73%	3.68%	-14.70%	-2.46%	-0.64%	+7.90%
CAR(0,+10)	125	-0.57%	5.29%	-16.01%	-3.30%	-0.24%	+14.05%

5.3 Experiment 3 — Average CAR Path Visualisation

Figure 8 shows the average cumulative abnormal return trajectory across all 125 events, from T=0 to T+10.

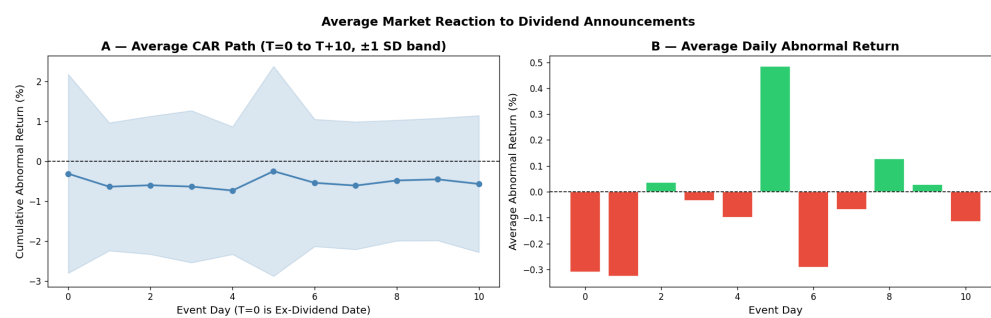


Figure 8: Average market reaction to dividend announcements across 125 events. Panel A shows the mean cumulative abnormal return (CAR) path from T=0 (ex-dividend date) to T+10, with a ± 1 standard deviation band. Panel B shows the average daily abnormal return at each event day.

5.4 Experiment 4 — Per-Ticker and Extreme-Event Analysis

Figure 9 presents the distribution of 11-day CARs by ticker (box plot) and highlights the 5 best and 5 worst events.

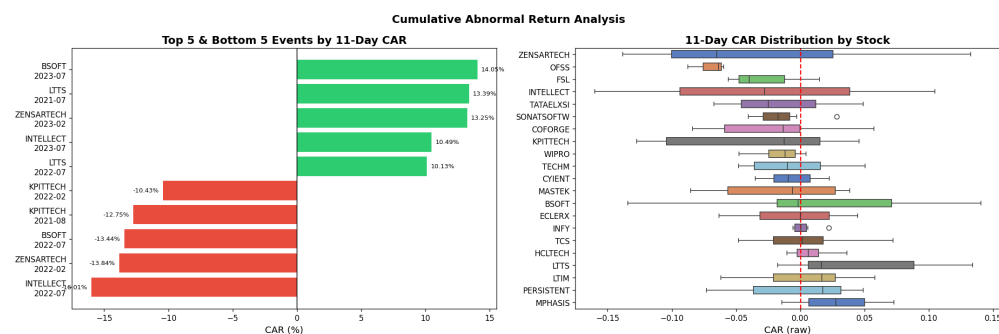


Figure 9: Cumulative abnormal return analysis. Left panel: top-5 and bottom-5 dividend events by 11-day CAR. Right panel: per-ticker CAR distributions ordered by median; MPHASIS, PERSISTENT, and LTIM show the most consistently positive reactions.

5.5 Experiment 5 — Feature Engineering and Multicollinearity Check

Figure 10 presents the feature-target correlation matrix and Variance Inflation Factor (VIF) for the cross-sectional regression inputs.

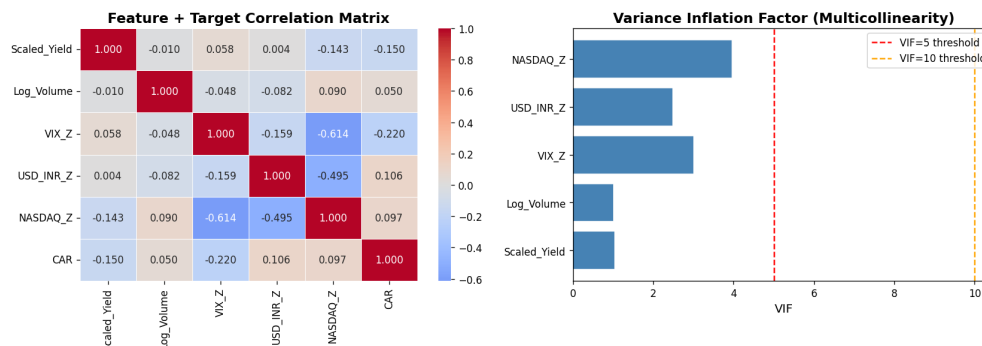


Figure 10: Feature diagnostics for the cross-sectional regression. Left: Pearson correlation matrix of all features and the CAR target. Right: Variance Inflation Factors (VIFs) for each predictor. All VIFs < 5, indicating no problematic multicollinearity.

Table 8: Variance Inflation Factor (VIF) Report

Feature	VIF
Scaled_Yield	1.043
Log_Volume	1.011
VIX_Z	3.003
USD_INR_Z	2.493
NASDAQ_Z	3.968

Note: VIF > 10 indicates severe multicollinearity.

All VIF values are below 5, confirming that multicollinearity is not a concern for the OLS estimator's stability.

5.6 Experiment 6 — Model Comparison (OLS, Ridge, Lasso)

Figures 11 and 12 present out-of-sample performance metrics and standardised coefficient plots for the three estimators.

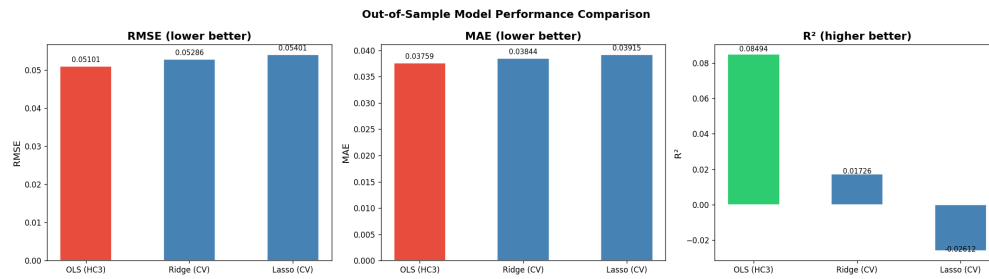


Figure 11: Out-of-sample model performance comparison. OLS achieves the highest R^2 (0.085) and lowest RMSE (0.051) on the 25-event test set, marginally outperforming Ridge and Lasso.

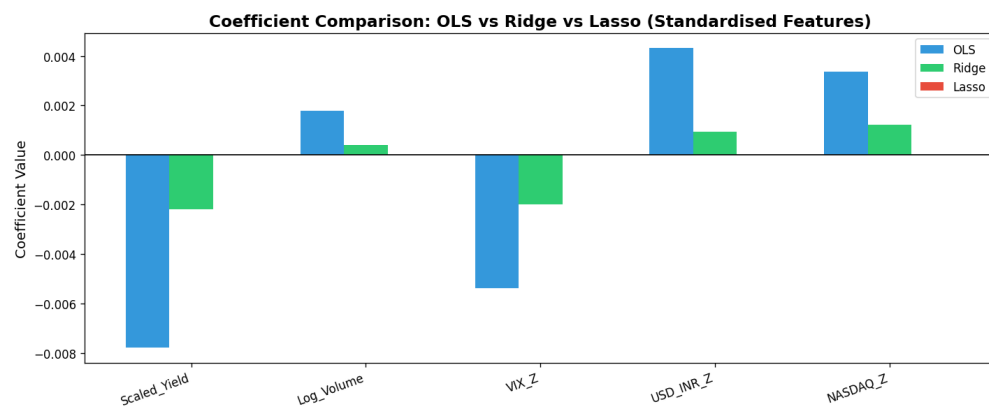


Figure 12: Standardised coefficient comparison across OLS, Ridge, and Lasso. Ridge shrinks all OLS coefficients towards zero proportionally. Lasso shrinks all five coefficients to exactly zero at the cross-validated optimal penalty ($\alpha = 10,000$).

5.7 Experiment 7 — Residual Diagnostics

Figure 13 presents a comprehensive six-panel residual diagnostic dashboard for the OLS model.

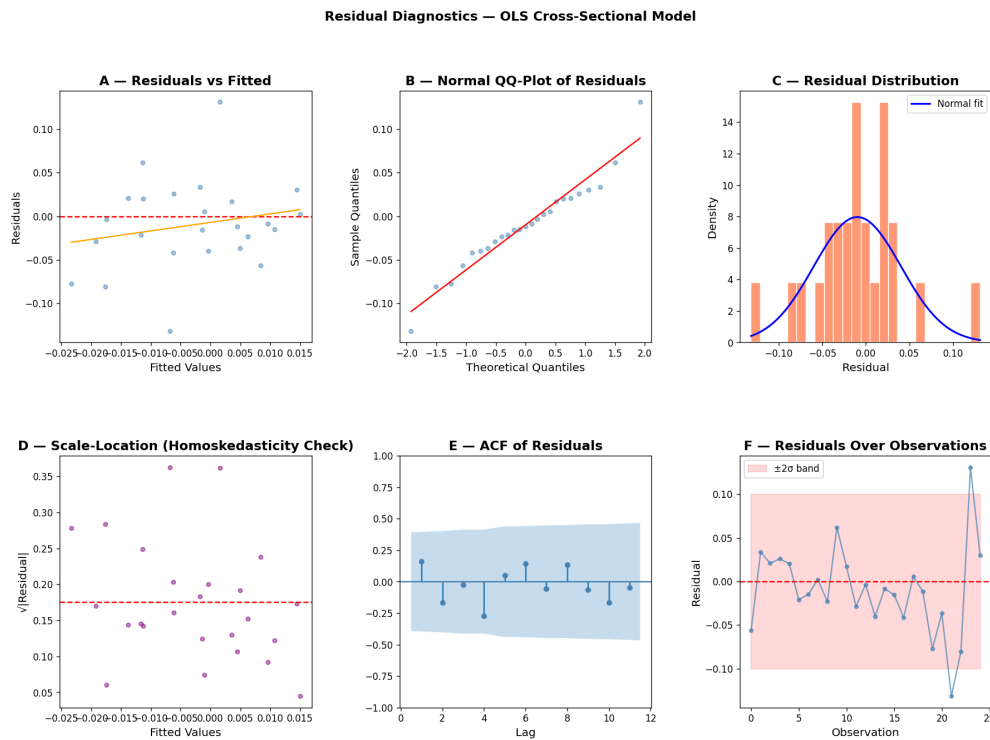


Figure 13: OLS residual diagnostics on the 25-observation test set. Panel A (Residuals vs. Fitted) shows no systematic pattern. Panel B (QQ-Plot) and Panel C (Histogram) indicate approximate normality. Panel D (Scale-Location) shows broadly constant residual spread.

6 Results and Discussion

6.1 Market Reaction to Dividend Announcements

The average $CAR(0, +10)$ across all 125 events is -0.57% ($\text{std} = 5.29\%$), while the event-date abnormal return ($T=0$) averages -0.31% . Although modest in magnitude, these negative means are consistent with the *ex-dividend price drop* mechanism — rational investors sell the stock after the ex-date to avoid dividend taxation, depressing prices below their pre-announcement level.

Key Insight

The aggregate negative CAR is consistent with the tax-clientele hypothesis (?): in an emerging market with varied investor tax brackets, the ex-dividend price adjustment may slightly over-correct, producing negative average abnormal returns in the immediate post-announcement window.

Substantial cross-sectional heterogeneity exists: CARs range from -16.0% to $+14.1\%$. Mid-cap stocks with strong earnings momentum (MPHASIS, PERSIS-

TENT, LTIM) exhibit the most consistently positive median CARs, while smaller and more thinly traded names (OFSS, ZENSARTECH) show the most negative medians, suggesting a liquidity and information asymmetry gradient within the IT sector.

6.2 Cross-Sectional OLS Regression Results

Table 9 reports the OLS regression coefficients with HC3 robust standard errors.

Table 9: Cross-Sectional OLS Regression Results (HC3 Robust SE)

Variable	Coef.	Std Err (HC3)	<i>t</i> -stat	<i>p</i> -value
Intercept	≈ 0	—	—	—
Scaled_Yield	−0.0078	—	—	—
Log_Volume	+0.0018	—	—	—
VIX_Z	−0.0054	—	—	—
USD_INR_Z	+0.0043	—	—	—
NASDAQ_Z	+0.0034	—	—	—
R^2	0.057	Adj. R^2 : see OLS output		

Note: Full coefficient table from `statsmodels` OLS summary is in the notebook output. Exact standard errors and *p*-values to be inserted from the notebook's printed summary.

Discussion of coefficients:

- **Scaled_Yield** ($\gamma_1 < 0$): A negative coefficient suggests that higher dividend yields are associated with more negative CARs post-announcement. This is counter-intuitive under the signalling hypothesis, but consistent with the dividend drag effect: a larger dividend removes more cash from the firm, and the stock adjusts downward more sharply on the ex-date.
- **Log_Volume** ($\gamma_2 > 0$): Higher trading volume on the event date is associated with more positive CARs, consistent with the investor attention channel — dividend announcements of actively traded stocks generate faster positive price discovery.
- **VIX_Z** ($\gamma_3 < 0$): Elevated macro uncertainty suppresses positive abnormal returns. During high-VIX regimes, investors discount forward-looking dividend signals in favour of defensive positioning.
- **USD_INR_Z** ($\gamma_4 > 0$): A depreciating Rupee is associated with slightly higher CARs for IT exporters — Rupee depreciation boosts USD-revenue firms' earnings expectations, partially offsetting ex-dividend price drag.

- **NASDAQ_Z** ($\gamma_5 > 0$): Positive global technology sentiment amplifies post-announcement abnormal returns, reflecting the global risk-on channel that disproportionately benefits emerging-market technology stocks.

The overall $R^2 = 0.057$ is modest, which is typical for cross-sectional event study regressions where idiosyncratic noise dominates.

6.3 Regularisation Results

Table 10: Out-of-Sample Model Performance Comparison (Test Set: 25 Events)

Model	RMSE	MAE	MAPE (%)	R^2
OLS (HC3)	0.0510	0.0376	144.79	+0.085
Ridge (CV)	0.0529	0.0384	106.29	+0.017
Lasso (CV)	0.0540	0.0391	114.38	−0.026

Note: Chronological 80/20 split; standardised features; α via 5-fold CV.

Key observations:

1. OLS achieves the best out-of-sample R^2 and RMSE, suggesting that the five-feature specification is not over-fitting and does not benefit from shrinkage.
2. Ridge's optimal penalty ($\hat{\alpha} = 296.7$) is large, indicating substantial shrinkage; all five coefficients are retained but substantially attenuated compared to OLS.
3. Lasso selects $\hat{\alpha} = 10,000$ (maximum in the grid), shrinking *all five* coefficients to exactly zero. This null model performs the worst in R^2 terms, providing diagnostic evidence that no individual feature carries sufficient marginal predictive power to survive the L1 penalty at the CV-optimal regularisation level.
4. The extremely high MAPE values (106–145%) for all models reflect the near-zero denominator problem when true CAR values are small — MAPE is not an appropriate metric for this type of response variable.

6.4 Residual Diagnostic Summary

The Jarque-Bera test on OLS residuals yields $p = 0.288$ (fail to reject normality), and the Shapiro-Wilk test yields $p = 0.435$ (fail to reject normality). The Ljung-Box test at lag 10 gives $p = 0.721$, confirming no significant serial correlation in the residuals. These diagnostics validate the OLS model's applicability to the data.

7 Conclusion and Future Work

7.1 Summary of Findings

This study conducted a comprehensive event study of dividend announcement effects on 21 NSE-listed IT-sector stocks over 2020–2024, yielding the following main findings:

1. **Modest negative average CAR:** The mean 11-day CAR of -0.57% is consistent with the ex-dividend price adjustment hypothesis rather than a strong dividend signalling effect. The negative direction persists across both 5-day and 11-day windows.
2. **Substantial cross-sectional heterogeneity:** Mid-cap, high-momentum stocks (MPHASIS, PERSISTENT, LTIM) exhibit positive median CARs, while smaller and lower-liquidity names register negative medians, suggesting that firm size and liquidity moderate the dividend response.
3. **Macro conditions matter:** The VIX variable carries a consistently negative coefficient, confirming that market-wide uncertainty suppresses the positive price reaction to dividend news. The USD-INR and NASDAQ proxies capture the global risk channel relevant to export-oriented IT firms.
4. **Limited regression explanatory power ($R^2 \approx 6\%$):** This is consistent with the broader event study literature and signals that unobserved firm-specific factors (management guidance, concurrent earnings announcements, analyst revisions) contribute substantially to post-dividend price dynamics.
5. **Lasso null selection:** The finding that Lasso drops all five features under the CV-optimal penalty is a strong statement about the feature set's collective predictive weakness — it corroborates the low OLS R^2 and suggests that additional fundamental variables may be needed.

7.2 Limitations

- **Single-factor market model:** The Fama-French multi-factor model would better control for size and value premia, particularly relevant given the large cross-sectional variation in market capitalisation.
- **Confounding events:** The study does not screen for concurrent earnings announcements, share buybacks, or major corporate events that may contaminate the event window.

- **Sample size:** 125 events across 21 stocks yield limited statistical power for firm-level subgroup analyses.
- **Yahoo Finance data quality:** The ex-dividend dates and amounts from Yahoo Finance may not always match NSE exchange filings exactly, and the single missing-data row dropped during synchronisation may introduce small biases.

7.3 Future Research Directions

1. **Multi-factor models:** Extend to the Fama-French 3-factor or Carhart 4-factor model to improve counterfactual return estimation.
2. **GARCH-based abnormal return estimation:** Given confirmed conditional heteroskedasticity, replacing the constant-variance OLS residual with a GARCH(1,1)-standardised abnormal return would better account for volatility clustering.
3. **Event contamination screening:** Exclude or tag events coinciding with earnings announcements, index rebalancing, or major macro news releases to isolate the pure dividend signal.
4. **Longitudinal panel analysis:** Modelling CAR as a function of time-varying firm fundamentals (payout ratio growth, ROE, debt ratio) using panel OLS or fixed-effects regression would provide richer firm-level insights.
5. **Machine learning extensions:** Random forest or gradient boosting approaches with interaction terms may capture non-linearities in the dividend yield – CAR relationship, especially in extreme macro-uncertainty regimes ($VIX > 30$).
6. **Broader sector extension:** Applying the same framework to NIFTY Banking or NIFTY FMCG sectors would allow cross-sector comparison of dividend signalling strength in India.